



Chemistry Ch 2.1: States of Matter

Chapter 2, Matter · kitab ke page 15 se 26

Pehle bunyad

Matter wo har cheez hai jis mein **mass** ho aur jo **jagah ghere**. Matter atoms se bana hai, aur atom hi har maddey ka building block hai.

Aur riyasatein (states) alag kyun hoti hain? Sirf **atoms ki tarteeb (arrangement)** ki wajah se.

State	Pehchan	Misalein
Solid	Shakl bhi mutayyan, volume bhi. Zarray bilkul paas paas.	Barf, heera, dhatein
Liquid	Volume mutayyan, magar shakl bartan wali.	Paani, tel, doodh, shehad
Gas	Na shakl mutayyan na volume. Zarray door door.	O ₂ , H ₂ , N ₂
Plasma	Bohat ionized gas, electrons alag ho jate hain, bijli chalti hai.	Sooraj, aasmani bijli, welding ka arc

Do exotic states, ye aksar bhool jati hain

BEC (Bose Einstein Condensate): absolute zero ke qareeb, yani **-273.14°C**, jahan saaray zarray ek hi quantum state mein aa jate hain. Misal: rubidium ke atoms.

Liquid Crystals: liquid aur solid ke **darmiyan**. Bahti liquid ki tarah hain magar tarteeb solid wali. Misal: LCD screens, yani aap ka mobile.

Table 2.1: teenon ka muqabla

Property	Gas	Liquid	Solid
Density	Low	High	High
Compressibility	Very	Moderate	Not
Fluidity	Can flow	Can flow	Cannot flow

Quiz

1. Name the **four** states of matter with one example of each, then name the **two** exotic states and say what makes each one special.
2. Copy Table 2.1, comparing density, compressibility and fluidity for gas, liquid and solid.